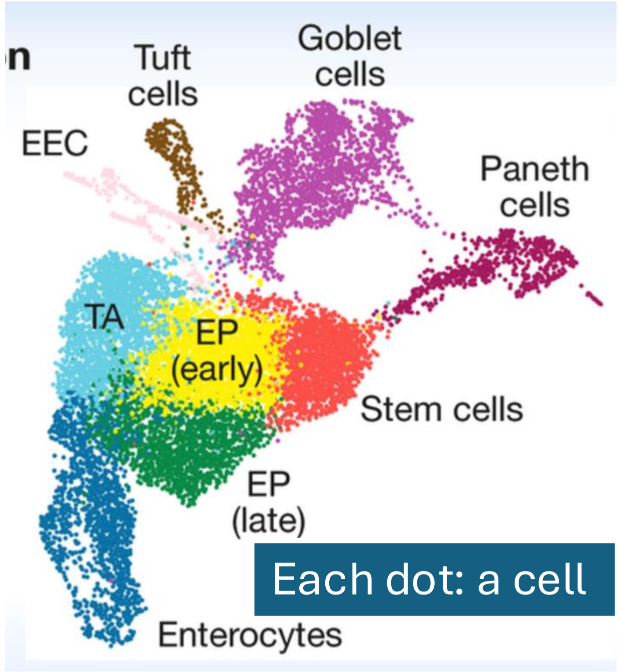
Single cell transcriptomics/ single cell RNA sequencing/ scRNAseq

Tissue

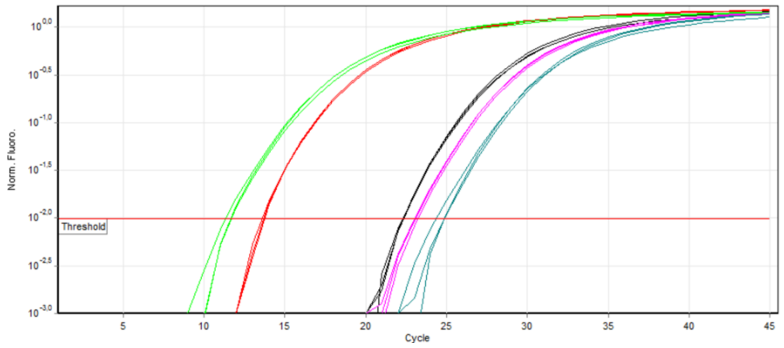
Ind. cells

Transcriptome sequencing

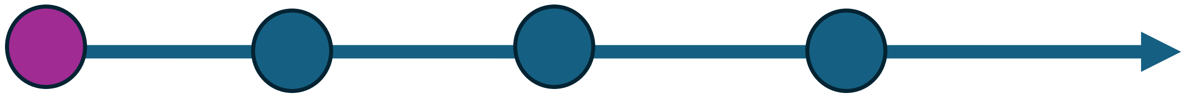


Luecken & Theis 2019

qRT-PCR

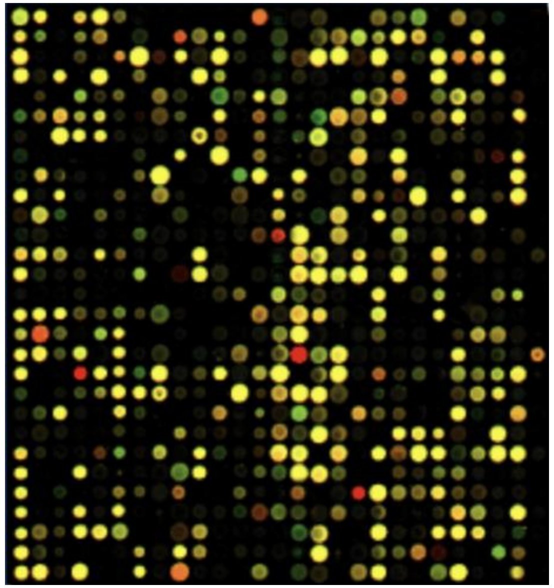


- PCR
- Sensitive
- A small number of genes/transcripts



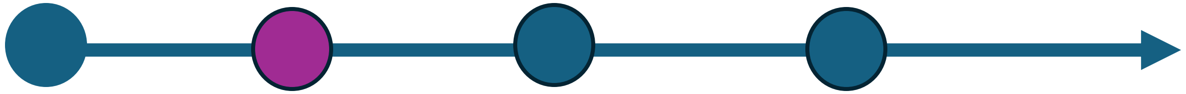
A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism

microarray

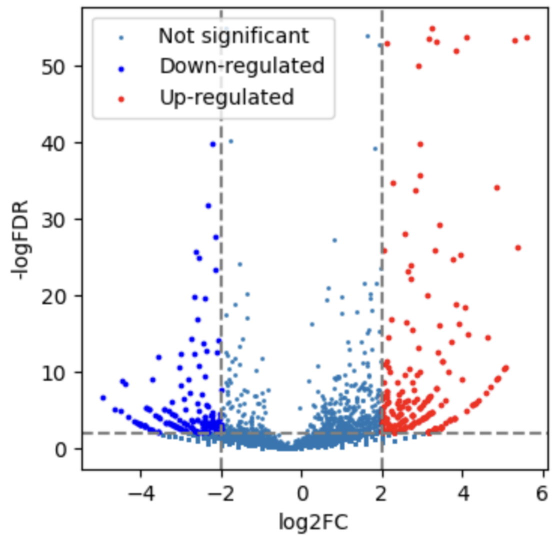


wikipedia

- Hybridization based
- Genome wide survey of the **transcriptome**
- Only profile predefined transcripts/genes on the chip



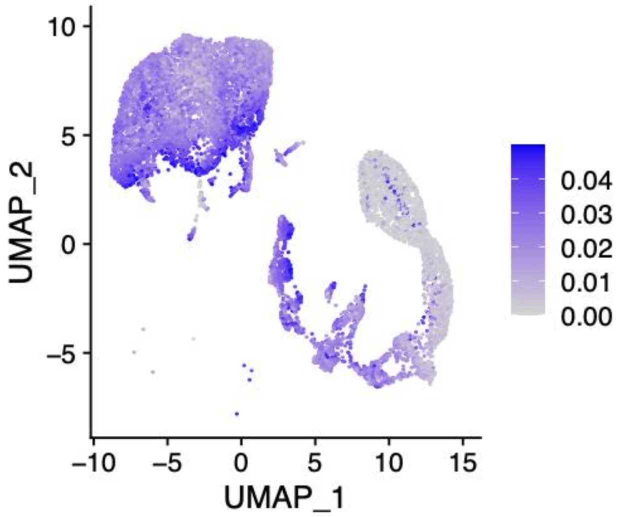
Bulk RNAseq



- Full sequencing of the whole **transcriptome**

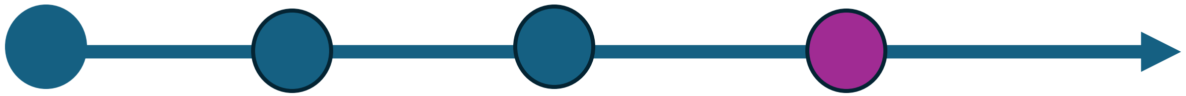
A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism

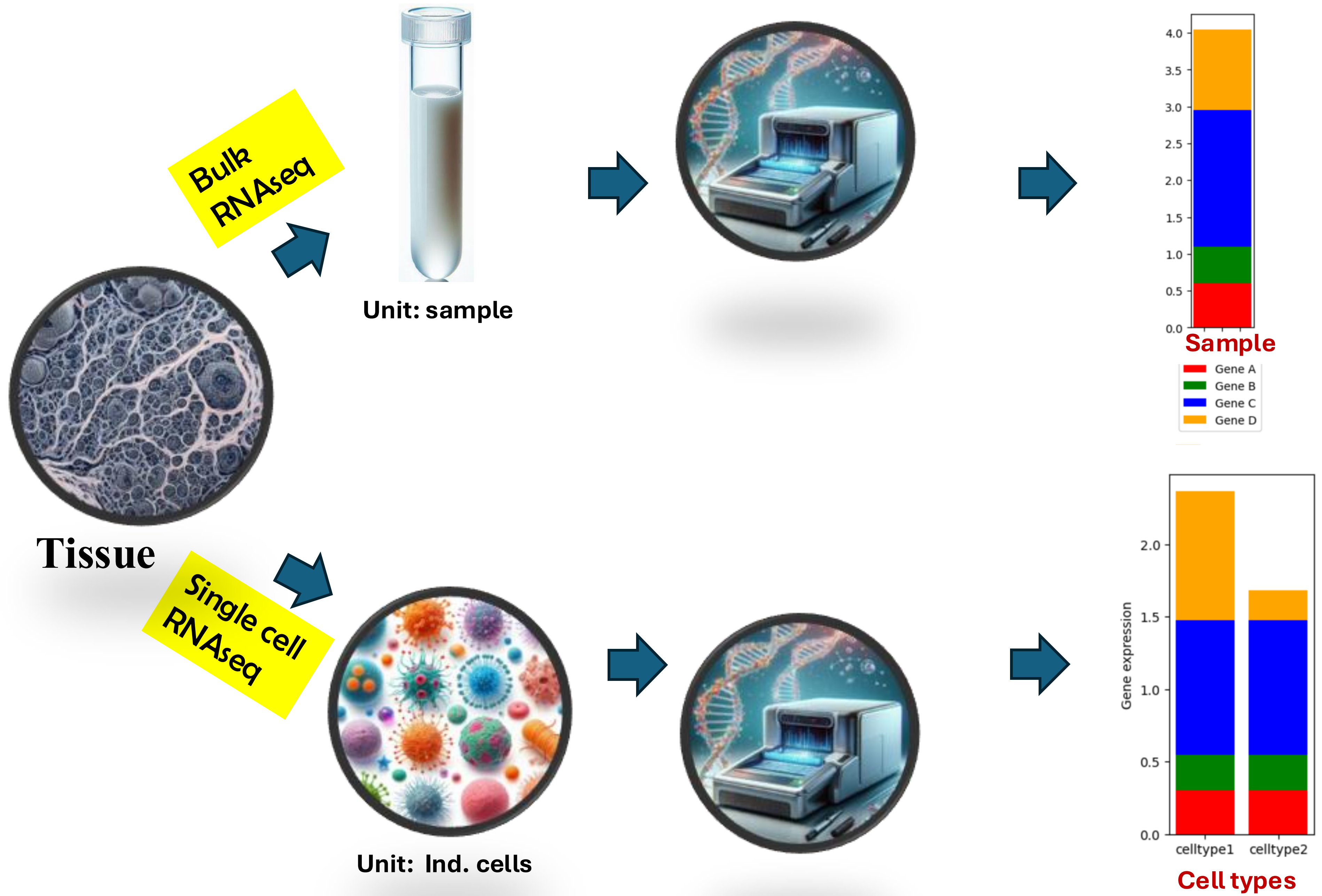
scRNAseq

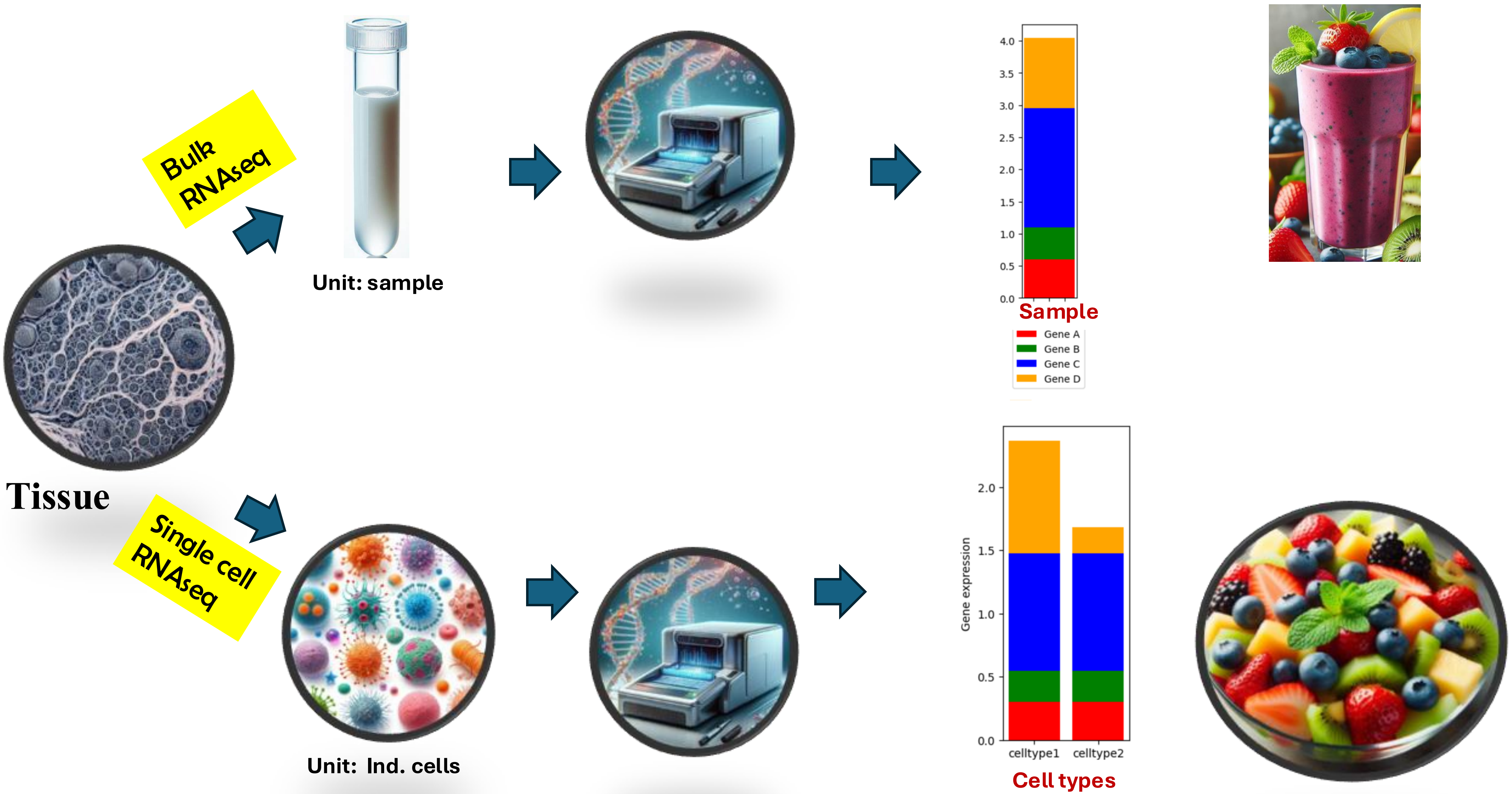


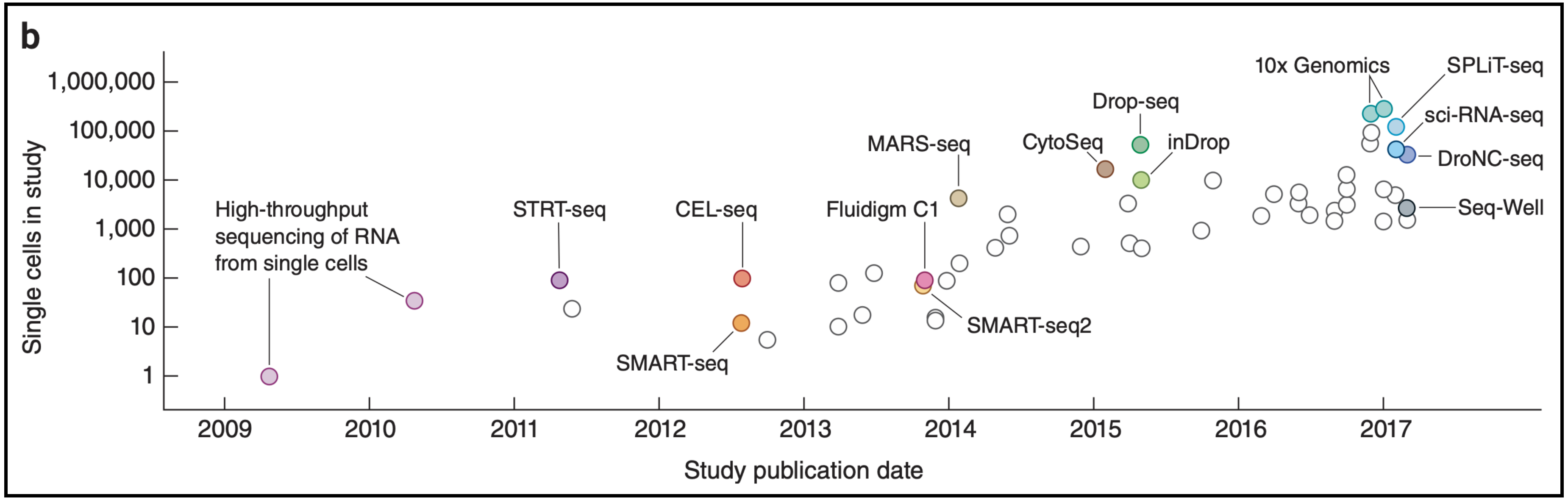
- Sequencing
- Genome wide survey of the **transcriptome**
- Individual cells

A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism

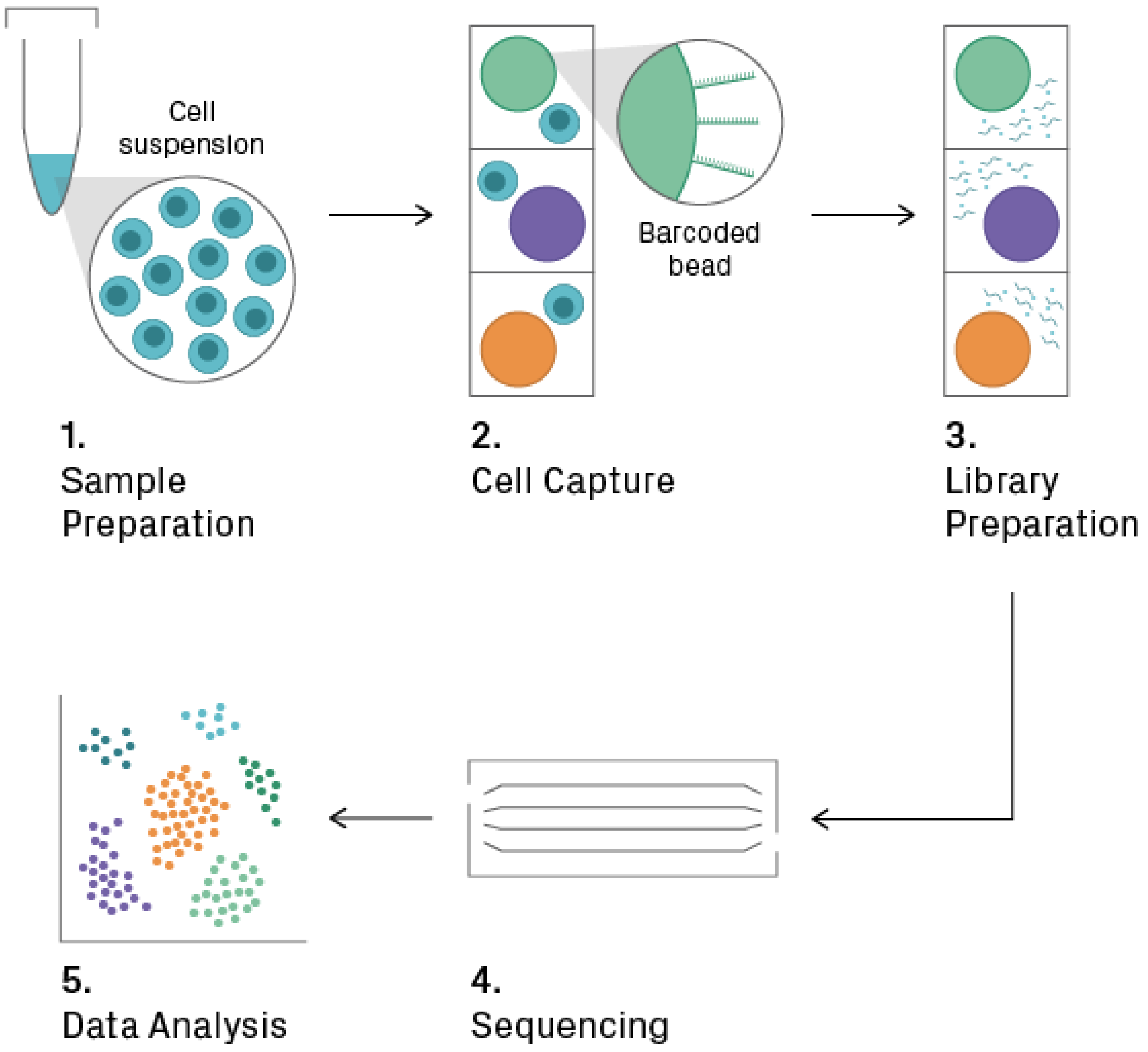



Bulk RNAseq
Unit: sample
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Sample
Gene A
Gene B
Gene C
Gene D
Tissue
Single cell RNAseq
Unit: Ind. cells
Gene expression
2.0
1.5
1.0
0.5
0.0
celltype1
celltype2
Cell types


Bulk RNAseq
Unit: sample
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Sample
Gene A
Gene B
Gene C
Gene D
Tissue
Single cell RNAseq
Unit: Ind. cells
Gene expression
2.0
1.5
1.0
0.5
0.0
celltype1
celltype2
Cell types

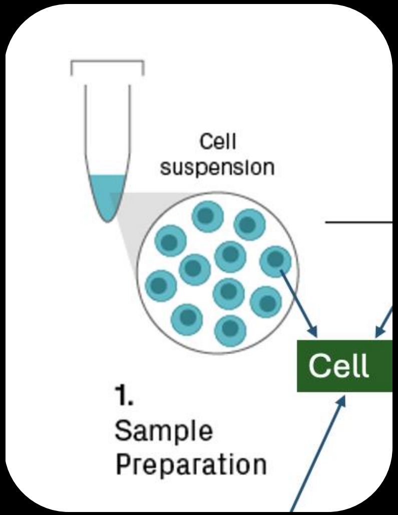


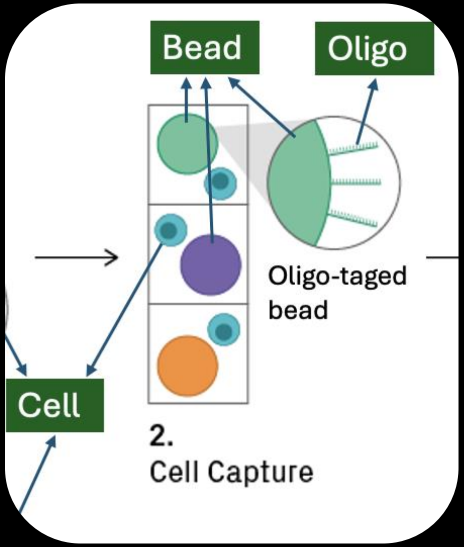
Svensson et al. 2018 Nature

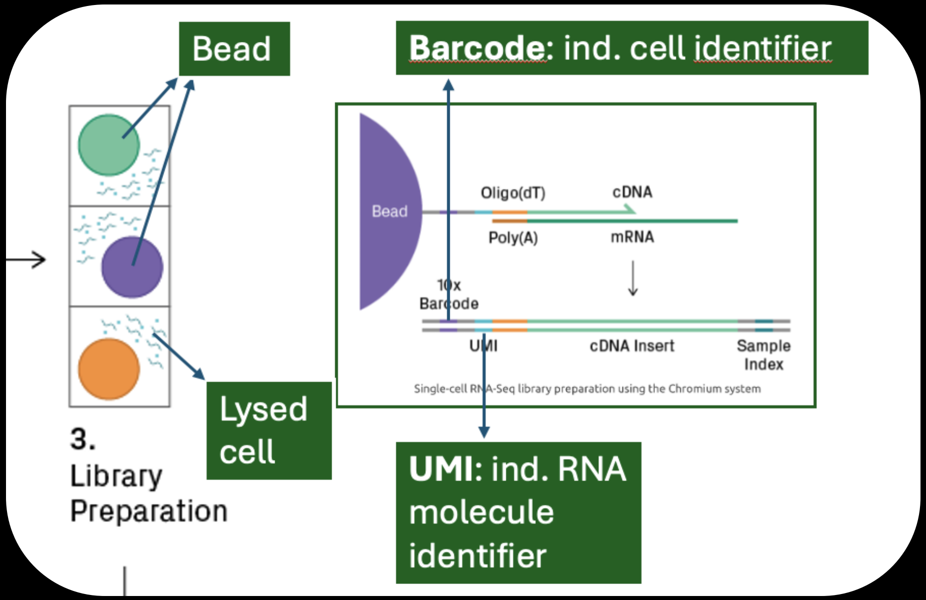


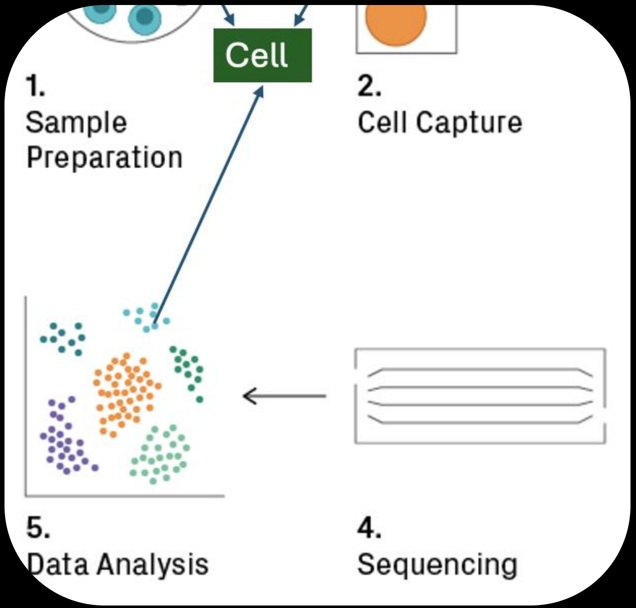
Single-cell sequencing workflow

https://www.azenta.com

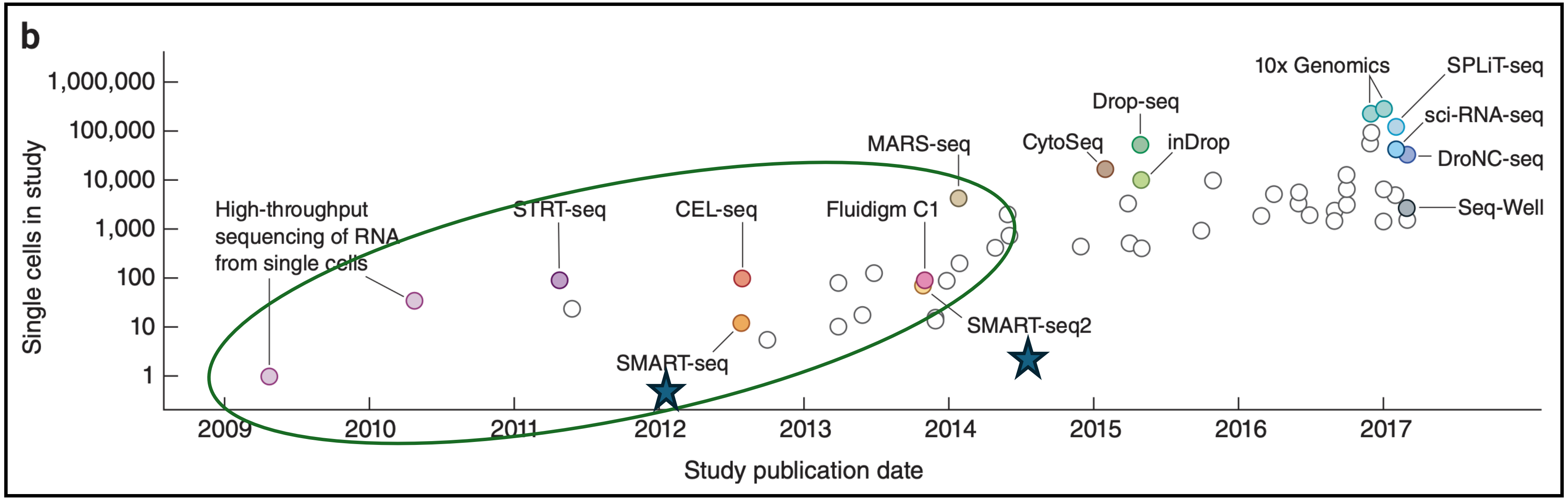

Cell
suspension
Cell
1.
Sample
Preparation


Bead
Oligo
Oligo-taged
bead
Cell
2.
Cell Capture


Bead
Barcode: ind. cell identifier
Bead
Oligo(dT)
cDNA
Poly(A)
mRNA
10x
Barcode
UMI
cDNA Insert
Sample
Index
Single-cell RNA-Seq library preparation using the Chromium system
3.
Library
Preparation
Lysed
cell
UMI: ind. RNA
molecule
identifier

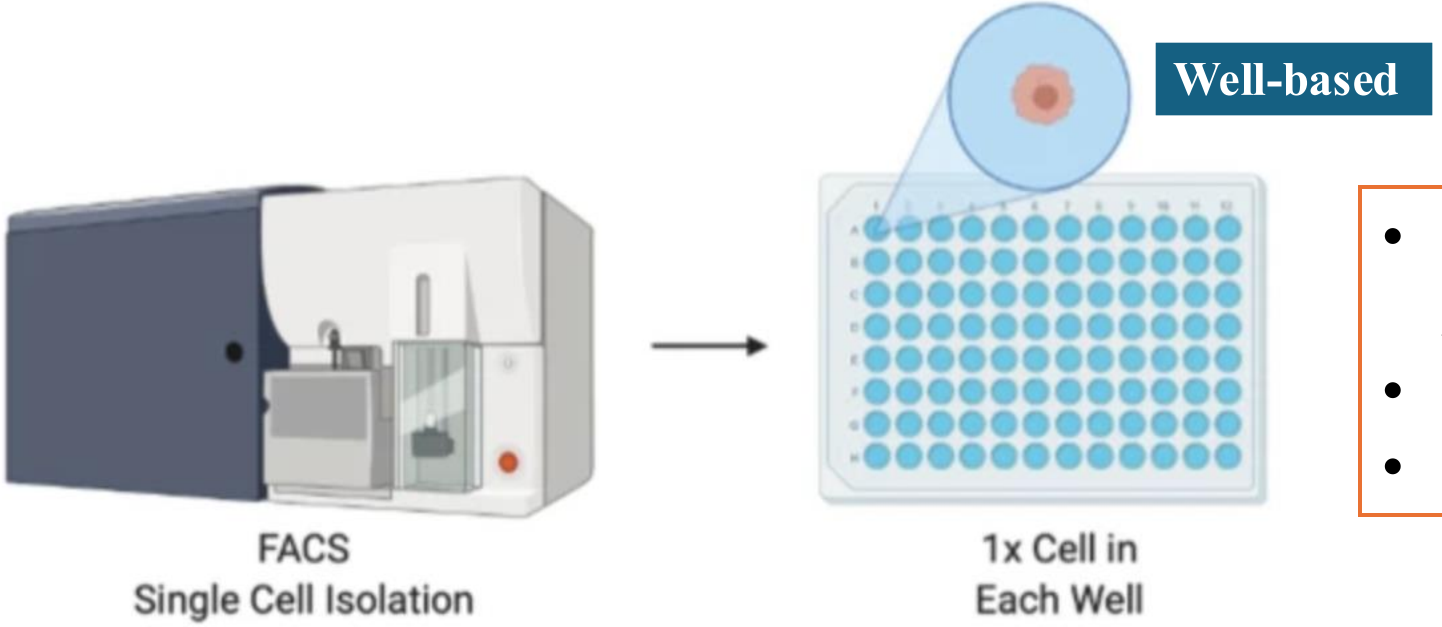


Cell
1.
Sample
Preparation
2.
Cell Capture
5.
Data Analysis
4.
Sequencing



Svensson et al. 2018 Nature

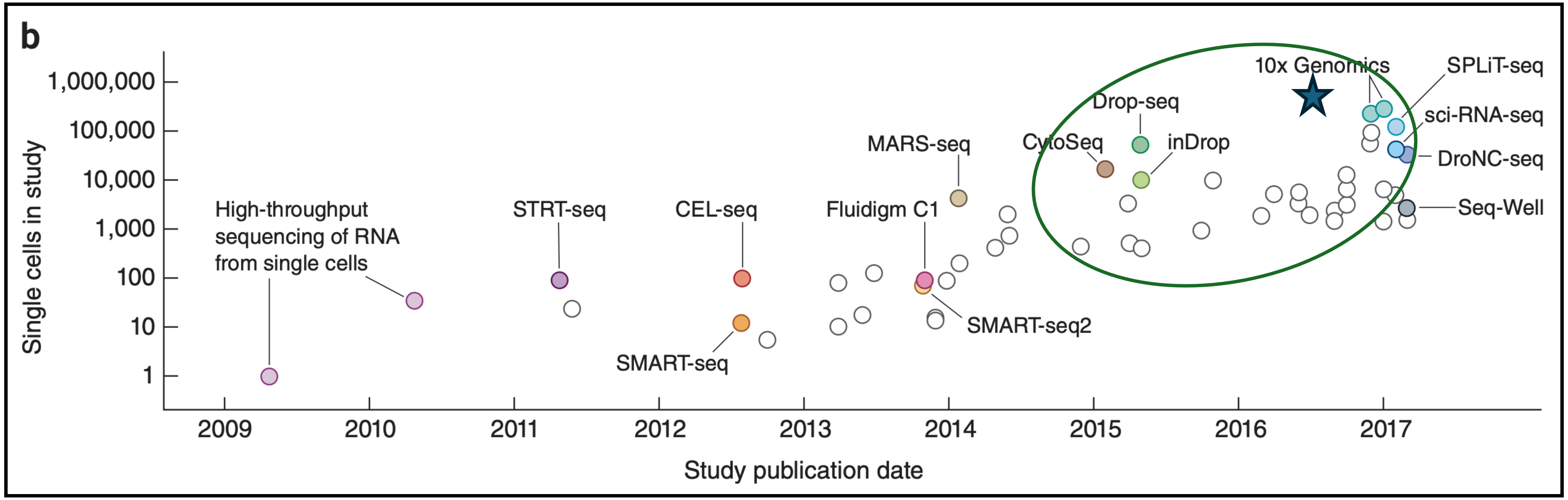
SMART-seq



- RNA full length sequencing /seq. deeper (isoform…)
- Less cells
- Lots of manual efforts

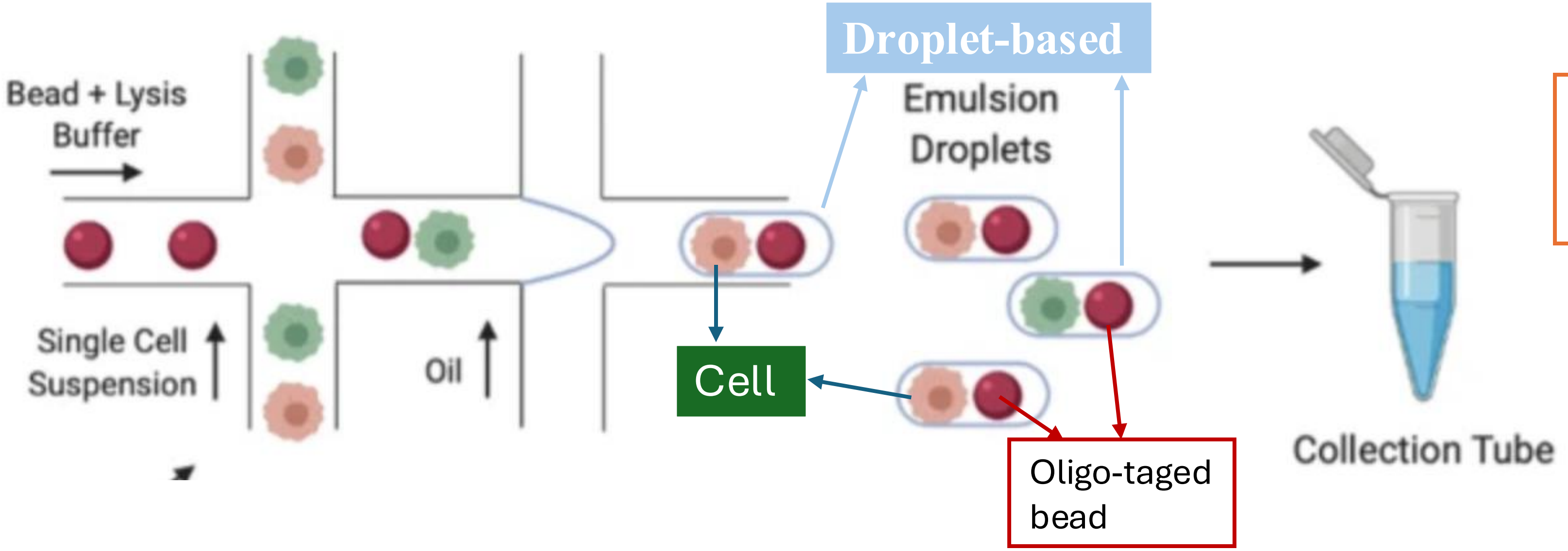
Flow cytometry

Probst et al. 2022 BMC genomics



Svensson et al. 2018 Nature

10X Genomics



- Lots of missing values
- A large number of cells

Microfluidic device

Probst et al. 2022 BMC genomics

Pair discussion

- Think about a situation where scRNAseq is better than bulk RNAseq

Learning goals

- ❖ The difference between bulk RNAseq and scRNAseq?